

Total No. of Questions : 8]

SEAT No. :

PE2233

[Total No. of Pages : 2

[6584]-134

B.E. (E & TC)

RADIATION AND MICROWAVE THEORY
(2019 Pattern) (Semester - VII) (404181)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of calculator is allowed.*
- 5) *Assume suitable data if necessary.*

- Q1) a)** Draw and explain the principle of operation of isolator. **[6]**
b) Draw Magic Tee and explain its working with the scattering matrix. **[6]**
c) Draw H plane Tee and explain its working with its scattering matrix. **[6]**

OR

- Q2) a)** Draw and explain working principle of two hole directional coupler with its scattering matrix. **[6]**
b) A 10 dBm directional coupler has a directivity of 40dBm, if the applied input power $P_1 = 10\text{mW}$, then calculate power outputs at port 2,3 and 4 of a lossless directional coupler. **[6]**
c) Draw E plane Tee and explain its working with its scattering matrix. **[6]**

- Q3) a)** Explain the operation of two cavity klystron with its constructional diagram. **[6]**
b) Write a short note on limitations and advantages of conventional tubes and solid-state devices. **[6]**
c) Draw and explain mode characteristics of single cavity klystron used in laboratory. Explain electronic tuning. **[6]**

OR

P.T.O.

- Q4)** a) Draw the construction of single cavity klystron tube and explain its working. [6]
b) A reflex klystron operates at the peak mode of $n = 2$ with beam voltage $V_0 = 300\text{V}$ and beam current $I_0 = 20\text{mA}$. If rf signal voltage $V_1 = 40\text{V}$, then determine the input power, output power and efficiency. [6]
c) Draw the construction and explain the working of Magnetron. [6]

- Q5)** a) Draw and explain V-I characteristics of Gunn diode to justify its amplifying behavior. [6]
b) Draw the construction and explain working of TRAPATT diode. [6]
c) Draw the construction and explain working of Varactor Diode. [5]

OR

- Q6)** a) Compare IMPATT, TRAPATT and BARITT diode with respect to its operating frequency, working principle, and efficiency. [6]
b) Explain construction and working principle of tunnel diode with its VI characteristics. [6]
c) Explain application of PIN diode as a modulator with neat diagrams. [5]

- Q7)** a) Explain any two applications of microwaves in detail. [5]
b) Explain impedance measurement using slotted transmission line. [6]
c) Explain the experimental set up to measure SWR and wavelength using slotted transmission line. [6]

OR

- Q8)** a) Define SAR, PEL and PET [6]
b) Explain attenuation measurement using power ratio measurement technique. State drawback of power ratio measurement technique. [5]
c) Explain any three health hazards due to radiations in details. [6]

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